Unit 8 ECOLOGY

Meet Chelsea Rochman, a pioneer and internationally recognized researcher on the ecological effects of microplastic pollution. After receiving a B.S. in Biology from University of California (UC) San Diego, Dr. Rochman completed a joint Ph.D. program in Ecology at UC Davis and San Diego State University. An Assistant Professor in the Department of Ecology and Evolutionary Biology at University of Toronto, Canada, Dr. Rochman and her students are studying how plastic waste and other contaminants affect individual organisms, populations, communities, and ecosystems. Dr. Rochman has also worked with industry and government leaders to shape science-based policy, including the Microbead-Free Waters Act in the United States and the G7 Ocean Plastics Charter led by Canada.

AN INTERVIEW WITH

Chelsea Rochman

How did you become interested in ecology and the impact of plastic pollutants?

I began my undergraduate studies at Santa Monica College, a community college near Los Angeles. Initially, I was really interested in the performing arts and the fine arts. But then I took a marine biology class, and I absolutely loved it. I switched to a biology major and after one year, I, transferred to UC San Diego. While there, I was able to study abroad, at the University of Queensland, Australia. While taking a course in ecology, we went to an island that was basically uninhabited. Even though there were so few people, there was debris all over the beach, and there were sea turtles harmed by plastic they had eaten. It was the first time I realized that there is a lot of plastic in the ocean and it was harming wildlife. I've never looked back—I've been working on this topic ever since.

What are microplastics, and what drew you to working on them?

Microplastics are small pieces of plastic debris, up to 5 mm in size. They are either produced as microplastics, such as microbeads in face wash, or they are fragments of larger plastic items, such as a bottle or bag. In 2009, I took my first trip to the Great Pacific Garbage Patch. But it didn't look like an island of trash the way it is described in the media. Instead, it is a soup of tiny plastic particles, the size of a pea or smaller. I realized that these small particles of plastic could infiltrate every level of the food chain. And so as I began my graduate work, I was focused on the small stuff because of what I had seen.

▼ Collecting samples of plastic waste.

How do plastic pollutants affect life in the ocean?

Plastic pollution is everywhere and affects marine life at every level of biological organization. Effects of microplastics can be complex, depending on their size and the type of polymer used to make them. Some microplastics cause liver tumors, others disrupt the endocrine system, and still others affect the number of offspring produced or their chances of survival. Large pieces of plastic also cause lots of problems. An animal can die after eating a big piece of plastic or being entangled by it. Large pieces also can spread diseases or smother sections of a coral reef, harming the community of organisms that live there. In addition, animals (or their eggs) attached to large pieces of plastic can be transported long distances, bringing those animals to new locations where they may harm existing communities.

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What is the most surprising discovery you've made?

The biggest surprise came gradually over my career: Microplastic waste is literally everywhere. People would ask, are microplastics in the food we eat or the water we drink? I was skeptical. The first time someone asked me about plastic in drinking water, I said, "We're not going to find anything." To check, I filtered some tap water and put the filter under a microscope. And of course we found it. I live near the Great Lakes, and we can find more than a hundred pieces of microplastic in one fish. It's in the air, tap water, bottled water, dust, the food we eat. And recently, we've learned that microplastics are in humans—they're present in human stool (feces). Now we're trying to learn how the microplastics we eat, drink, and breathe affect human health. Currently, we just don't know.

What can people do to reduce plastic pollution?

The first step is to grapple with the fact that plastic is a valuable resource. It's made from oil. It is not something that should be used as a throwaway material. Once we realize that, there are many things we can do. As individuals, we vote with our wallet when we buy things. We can refuse to buy single-use plastics, such as plastic straws, cups, or disposable razor blades. At the government level, we need to create a materials management stream where the plastic we use truly is recycled. Right now we use a lot of plastic products, and they go right into the trash, much of which ends up contaminating the environment. To fix this, we need local, national, and international agreements where we agree to reduce the amount of plastic that contaminates the environment by a certain amount, and we provide support and incentives to make that happen.